Constructing the Equivalent Homogeneous Rheology from a Multilayer Viscoelastic Model

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1 Overview

This folder contains the modular Wolfram Language pipeline for computing an equivalent homogeneous generalized Voigt rheology from an arbitrary N-layer viscoelastic body. The model is configured entirely through `ModelInput.wl` and is not hardcoded to any particular body.

The method follows Gevorgyan, Matsuyama & Ragazzo (2023, MNRAS 523, 1822; hereafter GMR 2023), which shows that the tidal Love number $k_2(s)$ of any N-layer Maxwell body can be exactly reproduced by a single homogeneous body with a generalized Voigt rheology. The layered-body formulation uses the propagator matrix notation of Sabadini, Vermeersen & Cambiotti (2016, hereafter SVC 2016).

The pipeline proceeds in three stages:

1. **Forward problem:** compute $k_2(s)$ as a rational function of the Laplace variable s using the propagator $\mathbf{\Pi}_\ell$ (SVC 2016, Eq. 2.9).

2. **Modal decomposition:** find the secular poles $\{s_j\}$ and residues $\{r_j\}$ via grid search on the secular determinant $D(s)$ (SVC 2016, Eq. 1.199), Brent refinement, and the limit formula.

3. **Rheological inversion:** map the poles and residues to the generalized Voigt compliance $C(s) = P_2(s)/P_1(s)$ (GMR 2023, Eq. 6) and extract parameters $(\gamma, \alpha, \eta, \alpha_i, \eta_i)$, then identify dominant modes for a reduced model.

2 File Inventory

File	Role
`ModelInput.wl`	User configuration: layer radii, densities, shear moduli, viscosities, tidal frequency, and numerical settings. Edit this file to change the body model.
`LayeredBodyConstants.wl`	Module 1: gravitational constant, gravity profile, Maxwell complex modulus, moment of inertia, mapping prefactor Θ, conversion factor F, formatting.
`YMatrix.wl`	Module 2: 6×6 fundamental matrix $\mathbf{Y}_\ell(r)$ and propagator $\mathbf{\Pi}_\ell(r, r')$ for degree $\ell = 2$ (SVC 2016, Eq. 2.42). Core–mantle boundary matrix $\mathbf{I}_C$ (Eq. 2.6).
`NLayerJacobian.wl`	Module 5: N-layer propagator using the unified 6×6 $\mathbf{\Pi}_\ell$ propagation through all layers (solid and liquid). Liquid layers are handled via $\hat{\mu} \approx 10^{-20}$ regularization. The secular determinant $D(s)$ (SVC 2016, Eq. 1.199) is computed from the 3×3 projected matrix. Provides `nLayerK2Num` (numerical $k_2(s)$), `nLayerDetNum` ($D(s)$), `nLayerResidues` (residues via limit formula), and perturbation utilities.

File	Role
`HomogeneousRheology.wl`	Module 6: `computeHomogeneous` — the main function. Calls the N-layer modules to find poles, compute residues, and extract all Voigt parameters. Returns an `Association` with 16 keys.
`RunHomogeneous.wl`	Driver script. Loads all modules and `ModelInput.wl`, calls `computeHomogeneous`, identifies dominant modes, prints tables, and generates comparison plots.
`VisualizeLayers.wl`	Standalone script that reads `ModelInput.wl` and draws a sector cross-section with numbered layers and a text listing of radii and thicknesses. Uses layer names from `"layerNames"` if provided, otherwise labels generically. Exports `output/layer_cross_section.pdf`. No pipeline modules required.

Table 1: Files in this folder.

2.1 Module dependency graph

```
RunHomogeneous.wl
  |-- ModelInput.wl
  |-- HomogeneousRheology`
  |     |-- LayeredBodyConstants`
  |     +-- NLayerJacobian`
  |           |-- LayeredBodyConstants`
  |           +-- YMatrix`

VisualizeLayers.wl
  +-- ModelInput.wl       (standalone, no modules)
```

2.2 Required load order

```
Get["LayeredBodyConstants.wl"];
Get["YMatrix.wl"];
Get["NLayerJacobian.wl"];
Get["HomogeneousRheology.wl"];
```

3 Quick Start

3.1 Prerequisites

- **Wolfram Engine** (free) or **Mathematica** (licensed), version 12.0 or later.
- `wolframscript` on your `PATH`.

3.2 Running

From the `homogeneous_model/` directory:

```
wolframscript -file RunHomogeneous.wl
```

This loads the model from `ModelInput.wl`, computes the equivalent homogeneous Voigt body, identifies dominant relaxation modes, prints all results, and generates four Love number comparison plots.

3.3 Changing the body model

Edit `ModelInput.wl`. All arrays must have the same length. Set $\mu < 1\,\mathrm{Pa}$ for liquid layers. Use exact rationals for maximum precision.

4 Configuration: `ModelInput.wl`

4.1 `$ModelConfig` — Physical model

```
$ModelConfig = <|
  "rk" -> {r1, r2, ..., rN}, (* outer radii, m *)
  "rhok" -> {rho1, ..., rhoN}, (* densities, kg/m^3 *)
  "muk" -> {mu1, ..., muN}, (* shear moduli, Pa *)
  "etak" -> {eta1, ..., etaN}, (* viscosities, Pa s *)
  "omega" -> 2 Pi / (T * 86400), (* tidal frequency, rad/s *)
  "layerNames" -> {"Layer␣1", ..., "Layer␣N"} (* optional *)
|>;
```

Key	Units	Description
`"rk"`	m	Outer radius of each layer, center to surface.
`"rhok"`	$\mathrm{kg/m^3}$	Density of each layer.
`"muk"`	Pa	Shear modulus. Set $\mu < 1\,\mathrm{Pa}$ for liquid layers.
`"etak"`	Pa s	Viscosity (any small value for liquid).
`"omega"`	rad/s	Tidal forcing frequency.
`"layerNames"`	—	Optional list of N strings for layer labels in `VisualizeLayers.wl`. If omitted, layers are labelled generically as "Layer i (solid/liquid)".

Table 2: `$ModelConfig` keys.

4.2 `$AnalysisConfig` — Numerical settings

5 Main Function: `computeHomogeneous`

```
computeHomogeneous[rk, rhok, muk, etak, omega, opts]
```

Key	Default	Description
`"workingPrecision"`	50	Digits of precision.
`"gridLogMin"`	-20	$\log_{10}\lvert s\rvert$ lower bound for pole search.
`"gridLogMax"`	-3	$\log_{10}\lvert s\rvert$ upper bound.
`"gridPoints"`	2000	Grid points for sign-change search.
`"omegaInterestMin"`	10^{-10}	Lower bound of frequency interest range (rad/s).
`"omegaInterestMax"`	10^{-5}	Upper bound (rad/s).
`"dominantThreshold"`	1/50	Fractional threshold for dominant-mode selection (2%).

Table 3: `$AnalysisConfig` keys.

5.1 Algorithm

1. **Elastic Love number.** Evaluate $k_E = \mathrm{Re}[k_2(10^{10})]$.
2. **Tidal k_2.** Evaluate $k_2(i\omega)$.
3. **Pole-finding.** Grid search for zeros of $D(s)$ (SVC 2016, Eq. 1.199) on the negative real s-axis via `nLayerDetNum`, detect sign changes, refine with Brent.
4. **Residues.** Limit formula $r_j = \delta \cdot k_2(s_j + \delta)$ with $\delta = 10^{-6}|s_j|$.
5. **Fluid Love number.** $k_f = k_E + \sum(-r_j/s_j)$ (GMR 2023, Eq. 26).
6. **Mapping constants.** $\Theta = 3\mathrm{I}_\circ G/R^5$ (GMR 2023, Eq. 2), $\gamma = \Theta/k_f$ (Eq. 4), $\alpha = \Theta/k_E - \gamma$, $F = 5\bar{\rho}R^2/38$.
7. **Voigt inversion.** Build compliance $C(s) = P_2(s)/P_1(s)$ (Eq. 6), factor $s = 0$ root from P_1 for η, solve $P_2(s) = 0$ for Voigt poles $\{q_i\}$ (Eq. 20), extract (α_i, η_i) from partial fractions of $C_v(s)$ (Eq. 23).

5.2 Return value

Returns a Wolfram `Association` with 16 keys:

Key	Units	Description
`"kE"`	dimless	Elastic Love number k_E.
`"kf"`	dimless	Fluid Love number k_f.
`"k2Re"`	dimless	$\mathrm{Re}[k_2(i\omega)]$.
`"k2Im"`	dimless	$\mathrm{Im}[k_2(i\omega)]$.
`"gamma"`	Pa	Gravitational rigidity γ (GMR 2023, Eq. 4).
`"alpha"`	Pa	Elastic rigidity α.
`"eta"`	Pa s	Isolated dashpot η.
`"prefactor"`	s^{-2}	$\Theta = 3\mathrm{I}_\circ G/R^5$ (GMR 2023, Eq. 2).
`"Fconv"`	$\mathrm{Pa\,s}^2$	Conversion factor F.
`"nPoles"`	int	Number of secular poles M.

Key	Units	Description
`"poles"`	1/s	Pole locations $\{s_j\}$.
`"residues"`	dimless	Residues $\{r_j\}$.
`"nVoigt"`	int	Number of Voigt elements N_V.
`"tauV"`	s	Relaxation times $\tau_i = -1/p_i$.
`"alphaV"`	Pa	Voigt spring constants α_i.
`"etaV"`	Pa s	Voigt dashpot viscosities η_i.

Table 4: Return value keys of `computeHomogeneous`.

6 Driver Output: `RunHomogeneous.wl`

The driver prints seven output sections:

1. **Love numbers:** k_E, k_f, $\mathrm{Re}[k_2]$, $\mathrm{Im}[k_2]$.
2. **Rheological parameters:** γ, α, η, pole/element counts.
3. **Secular relaxation modes table:** all M poles with τ_j, residues, sorted by decreasing $|s_j|$.
4. **Voigt elements table:** all N_V elements with τ_i, α_i, η_i, sorted by decreasing τ.
5. **Dominant mode analysis:** fractional contribution of each element to $-\mathrm{Im}[J]$ over the interest range.
6. **Reduced model summary:** k_2 from dominant-only model with relative errors.
7. **Comparison plots:** $-\mathrm{Im}[k_2]$ and $\mathrm{Re}[k_2]$ vs ω: N-layer vs full Voigt, N-layer vs reduced Voigt.

7 Mathematical Pipeline

7.1 From multilayer to $k_2(s)$

Each solid layer has Maxwell rheology with complex shear modulus (SVC 2016, Correspondence Principle):

$$\hat{\mu}_k(s) = \frac{\mu_k\,\nu_k\,s}{\mu_k + \nu_k\,s}\,, \tag{1}$$

where μ_k is the shear modulus and ν_k the viscosity (`"etak"` in the code). Liquid layers use the regularization $\hat{\mu} \approx 10^{-20}$ Pa, which makes the 6×6 fundamental matrix $\mathbf{Y}_\ell(r)$ (SVC 2016, Eq. 2.42) nearly rank-5 (the S row vanishes). All layers are propagated through the full 6×6 propagator $\mathbf{\Pi}_\ell(r, r') = \mathbf{Y}_\ell(r)\,\mathbf{Y}_\ell^{-1}(r')$ (Eq. 2.9) chain from the core–mantle boundary matrix $\mathbf{I}_C$ (Eq. 2.6); no special Clairaut propagator is needed. This yields $k_2(s)$ as a rational function of s for arbitrary layer configurations (any combination of solid and liquid layers).

7.2 Partial fraction decomposition

The poles $s_j < 0$ of $k_2(s)$ are the secular relaxation modes, i.e. the roots of the secular determinant $D(s) = 0$ (SVC 2016, Eq. 1.200). The partial-fraction form is (GMR 2023, Eq. 26):

$$k_2(s) = k_E + \sum_{j=1}^{M} \frac{r_j}{s - s_j}\,, \tag{2}$$

where r_j are the residues and $\tau_j = -1/s_j$ the relaxation times of the normal modes.

7.3 Compliance inversion

The compliance is written as a ratio of two polynomials (GMR 2023, Eq. 6):

$$C(s) = \frac{P_2(s)}{P_1(s)} = \frac{k_2(s)}{\Theta - \gamma\, k_2(s)}\,, \tag{3}$$

where $\Theta = 3\mathrm{I}_\circ G/R^5$ (Eq. 2) and $\gamma = \Theta/k_f$ is the gravitational rigidity (Eq. 4). The viscous part of the compliance decomposes into partial fractions as the generalized Voigt form (Eq. 23):

$$C_v(s) = \frac{\alpha^2}{(\gamma+\alpha)^2}\left(\frac{1}{\eta} + \sum_{i=1}^{n} \frac{1}{\eta_i}\right)\left(\frac{A_1}{s - s_1} + \cdots + \frac{A_{n+1}}{s - s_{n+1}}\right), \tag{4}$$

from which the full compliance is (Eq. 14):

$$C(s) = \frac{1}{\alpha + \gamma} + C_v(s)\,. \tag{5}$$

The isolated dashpot η and the Kelvin–Voigt element parameters α_i, η_i (with characteristic times $\omega_i^{-1} = \eta_i/\alpha_i$, Eq. 18) are extracted from the partial-fraction coefficients. Physical (Pa) values: multiply by $F = 5\bar{\rho}R^2/38$.

7.4 Dominant mode selection

Each Voigt element's maximum fractional contribution to $-\operatorname{Im}[C(i\omega)]$ is computed over $[\omega_{\min}, \omega_{\max}]$. Elements exceeding the threshold are retained in the reduced model.

8 Customising for a Different Body

1. Edit `ModelInput.wl`: set `"rk"`, `"rhok"`, `"muk"`, `"etak"` for your body. All arrays must have the same length N.
2. Set `"omega"` to the body's tidal forcing frequency.
3. Adjust `"gridLogMin"`/`"gridLogMax"` if relaxation times span a different range than the default.
4. Adjust `"omegaInterestMin"`/`"omegaInterestMax"` to the frequency range relevant for your problem.
5. Run: `wolframscript -file RunHomogeneous.wl`.

The pipeline auto-detects the number of layers, identifies liquid layers ($\mu < 1\,\mathrm{Pa}$), and adjusts the propagator chain accordingly. No code changes are needed.

9 Test Cases

The `test_cases/` folder contains input files for three tidally-heated bodies beyond the default 5-layer Moon model. These validate the pipeline for different layer counts and liquid-layer patterns.

`ModelInputEuropa.wl` — 4 layers (liquid iron core, silicate mantle, subsurface ocean, ice shell). Two liquid layers. Tidal period 3.5512 days. Based on Wahr et al. (2006), Hussmann & Spohn (2004).

`ModelInputEnceladus.wl` — 3 layers (silicate core, subsurface ocean, ice shell). One liquid layer. Tidal period 1.3702 days. Based on Cadek et al. (2016), Thomas et al. (2016).

`ModelInputIo.wl` — 4 layers (liquid Fe-FeS core, mantle, asthenosphere, lithosphere). One liquid layer, contrasting solid viscosities. Tidal period 1.7691 days. Based on Segatz et al. (1988), Bierson & Nimmo (2016).

9.1 Running a test case

Copy the desired input file over `ModelInput.wl` and run the pipeline:

```
cd homogeneous_model/
cp test_cases/ModelInputEuropa.wl ModelInput.wl
wolframscript -file RunHomogeneous.wl
wolframscript -file VisualizeLayers.wl
```

Repeat for Enceladus or Io by substituting the file name. Remember to restore the original `ModelInput.wl` afterwards if needed.

10 Layer Visualization: `VisualizeLayers.wl`

Standalone script that generates a sector cross-section of the layered body. Works for any number of layers.

```
wolframscript -file VisualizeLayers.wl
```

Input. Reads `ModelInput.wl` (`"rk"`, `"muk"`, and optionally `"layerNames"`).

Liquid detection. $\mu < 1\,\mathrm{Pa} \Rightarrow$ liquid.

Colours. Layers are coloured with the `LightTemperatureMap` palette: warm (orange) at the center, cool (blue) at the surface. The range is compressed to $[0.2, 0.8]$ to avoid extreme saturation.

Layer names. If `"layerNames"` is provided in `$ModelConfig`, those names are used directly. Otherwise, layers are labelled generically as "Layer i (solid)" or "Layer i (liquid)" based on the liquid detection. The pre-configured test cases include descriptive names (e.g. "Inner core", "Silicate mantle", "Ice shell").

Thin-layer exaggeration. If any layer occupies less than 3% of the surface radius, its visual thickness is boosted to the minimum and the deficit is redistributed proportionally from thicker layers. The title is updated to note "(visual thickness exaggerated)" when this occurs; otherwise it reads "(to scale)".

Layout. The cross-section sector is on the left with layer numbers placed along the left radial edge at each layer's midpoint, and "0" at the body centre. On the right, a text listing shows each layer as: `i --- Name  thickness km (radius km)`.

Output. `output/layer_cross_section.pdf` — sector cross-section with numbered layers on the left, text listing of names, thicknesses, and outer radii on the right.

11 Troubleshooting

Missing poles: Widen `"gridLogMin"`/`"gridLogMax"` or increase `"gridPoints"`.

Curves don't overlap in full Voigt plot: Increase `"workingPrecision"` (try 80 or 100).

`FindRoot` **warnings:** Usually harmless. If poles look wrong, increase grid density.

`NSolve::precw`**:** The Voigt pole solver needed extra precision; handled automatically at doubled precision.

References.

Gevorgyan, Matsuyama & Ragazzo (2023), MNRAS 523, 1822.

Sabadini, Vermeersen & Cambiotti (2016), Global Dynamics of the Earth: Applications of Viscoelastic Relaxation Theory to Solid-Earth and Planetary Geophysics, 2nd ed., Springer.